

# SHETH L.U.J AND SIR M.V COLLEGE

## PRACTICAL NO :- 12

AIM :- Combining datasets vertically (concatenation) using rbind() (R).

INPUT :-

```
data(iris)
```

```
spotify_data.clean <- read.csv("C:/Saurav S084/Adv python for data science/spotify_data clean.csv")
```

```
print("--- Data Structure Before Transformation ---")
```

```
print(names(iris))
```

```
print(names(spotify_data.clean))
```

```
iris_clean <- iris[, c("Species", "Sepal.Length")]
```

```
names(iris_clean) <- c("Species", "Height")
```

```
flower_clean <- spotify_data.clean[, c("artist_name", "track_duration_min")]
```

```
names(flower_clean) <- c("Species", "Height")
```

```
iris_clean$Height <- as.numeric(iris_clean$Height)
```

```
flower_clean$Height <- as.numeric(flower_clean$Height)
```

```
combined_data <- rbind(iris_clean, flower_clean)
```

```
print("--- Combined Data Summary ---")
```

```
print(paste("Iris rows:", nrow(iris_clean)))
```

```
print(paste("Spotify rows:", nrow(flower_clean)))
```

```
print(paste("Total rows (Expected):", nrow(iris_clean) + nrow(flower_clean)))
```

```
print(paste("Total rows (Actual):", nrow(combined_data)))
```

```
print("--- Preview of Combined Data (Top and Bottom) ---")
```

```
print(head(combined_data))
```

NAME :- SAURAV KANDARI

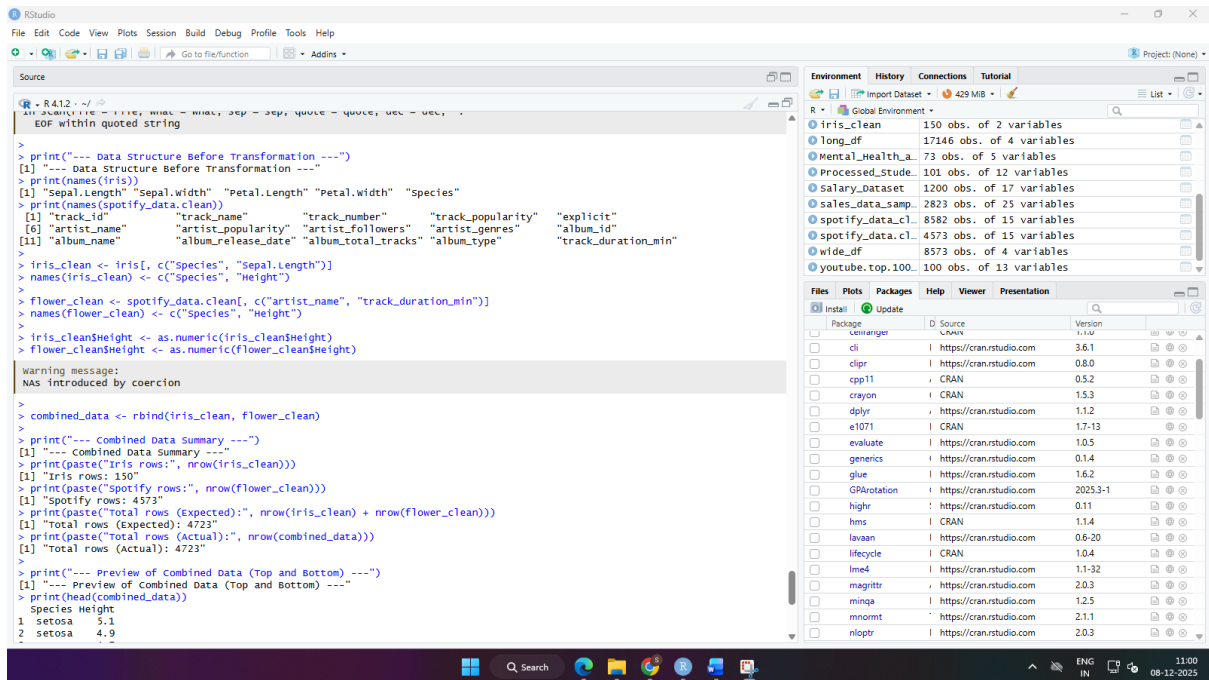
ROLL NO :- S084

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```
print(tail(combined_data))
```

OUTPUT :-



The screenshot shows the RStudio interface. The Source pane on the left contains R code for data cleaning and combining. The Environment pane on the right shows the objects created in the workspace. The console at the bottom shows the output of the code.

```
> print("---- data structure Before Transformation ----")
[1] "---- data structure Before Transformation ----"
> print(names(iris))
[1] "Sepal.Length" "Petal.Length" "Petal.Width" "Species"
> print(names(spotify_data_clean))
[1] "track_id" "track_name" "track_number" "track_popularity" "explicit"
[6] "artist_name" "artist_popularity" "artist_followers" "artist_genres" "album_id"
[11] "album_name" "album_release_date" "album_total_tracks" "album_type" "track_duration_min"
>
> iris_clean <- iris[, c("Species", "Sepal.Length")]
> names(iris_clean) <- c("Species", "Height")
>
> flower_clean <- spotify_data_clean[, c("artist_name", "track_duration_min")]
> names(flower_clean) <- c("Species", "Height")
>
> iris_clean$Height <- as.numeric(iris_clean$Height)
> flower_clean$Height <- as.numeric(flower_clean$Height)
warning message:
NAS introduced by coercion
>
> combined_data <- rbind(iris_clean, flower_clean)
>
> print("---- Combined Data Summary ----")
[1] "---- Combined Data Summary ----"
> print(paste("Iris rows:", nrow(iris_clean)))
[1] "Iris rows: 150"
> print(paste("Spotify rows:", nrow(flower_clean)))
[1] "Spotify rows: 4573"
> print(paste("Total rows (Expected):", nrow(iris_clean) + nrow(flower_clean)))
[1] "Total rows (Expected): 4723"
> print(paste("Total rows (Actual):", nrow(combined_data)))
[1] "Total rows (Actual): 4723"
>
> print("---- Preview of Combined Data (Top and Bottom) ----")
[1] "---- Preview of Combined Data (Top and Bottom) ----"
> print(head(combined_data))
Species Height
1 setosa 5.1
2 setosa 4.9
```

**Environment Pane:**

| Object          | Class      | Attributes                |
|-----------------|------------|---------------------------|
| iris_clean      | data.frame | 150 obs. of 2 variables   |
| long_df         | data.frame | 17146 obs. of 4 variables |
| Mental_Health_a | data.frame | 73 obs. of 3 variables    |
| Processed_Stude | data.frame | 101 obs. of 12 variables  |
| Salary_Dataset  | data.frame | 1200 obs. of 17 variables |
| sales_data_samp | data.frame | 2823 obs. of 25 variables |
| spotify_data_cl | data.frame | 8582 obs. of 15 variables |
| spotify_data.cl | data.frame | 4573 obs. of 15 variables |
| wide_df         | data.frame | 8573 obs. of 4 variables  |
| youtube.top.100 | data.frame | 100 obs. of 13 variables  |

**Files Pane:**

| Package     | Source                   | Version |
|-------------|--------------------------|---------|
| cli         | https://cran.rstudio.com | 3.6.1   |
| clipr       | https://cran.rstudio.com | 0.8.0   |
| cpp11       | CRAN                     | 0.5.2   |
| crayon      | CRAN                     | 1.5.3   |
| dplyr       | https://cran.rstudio.com | 1.1.2   |
| e1071       | CRAN                     | 1.7-13  |
| evaluate    | https://cran.rstudio.com | 1.0.5   |
| generics    | https://cran.rstudio.com | 0.1.4   |
| glue        | https://cran.rstudio.com | 1.6.2   |
| GPArotation | https://cran.rstudio.com | 20253-1 |
| highr       | https://cran.rstudio.com | 0.11    |
| hms         | CRAN                     | 1.1.4   |
| lavaan      | https://cran.rstudio.com | 0.6-20  |
| lifecycle   | CRAN                     | 1.0.4   |
| lime4       | https://cran.rstudio.com | 1.1-32  |
| magrittr    | https://cran.rstudio.com | 2.0.3   |
| minqa       | https://cran.rstudio.com | 1.2.5   |
| mnormt      | https://cran.rstudio.com | 2.1.1   |
| nloptr      | https://cran.rstudio.com | 2.0.3   |

NAME :- SAURAV KANDARI  
ROLL NO :- S084